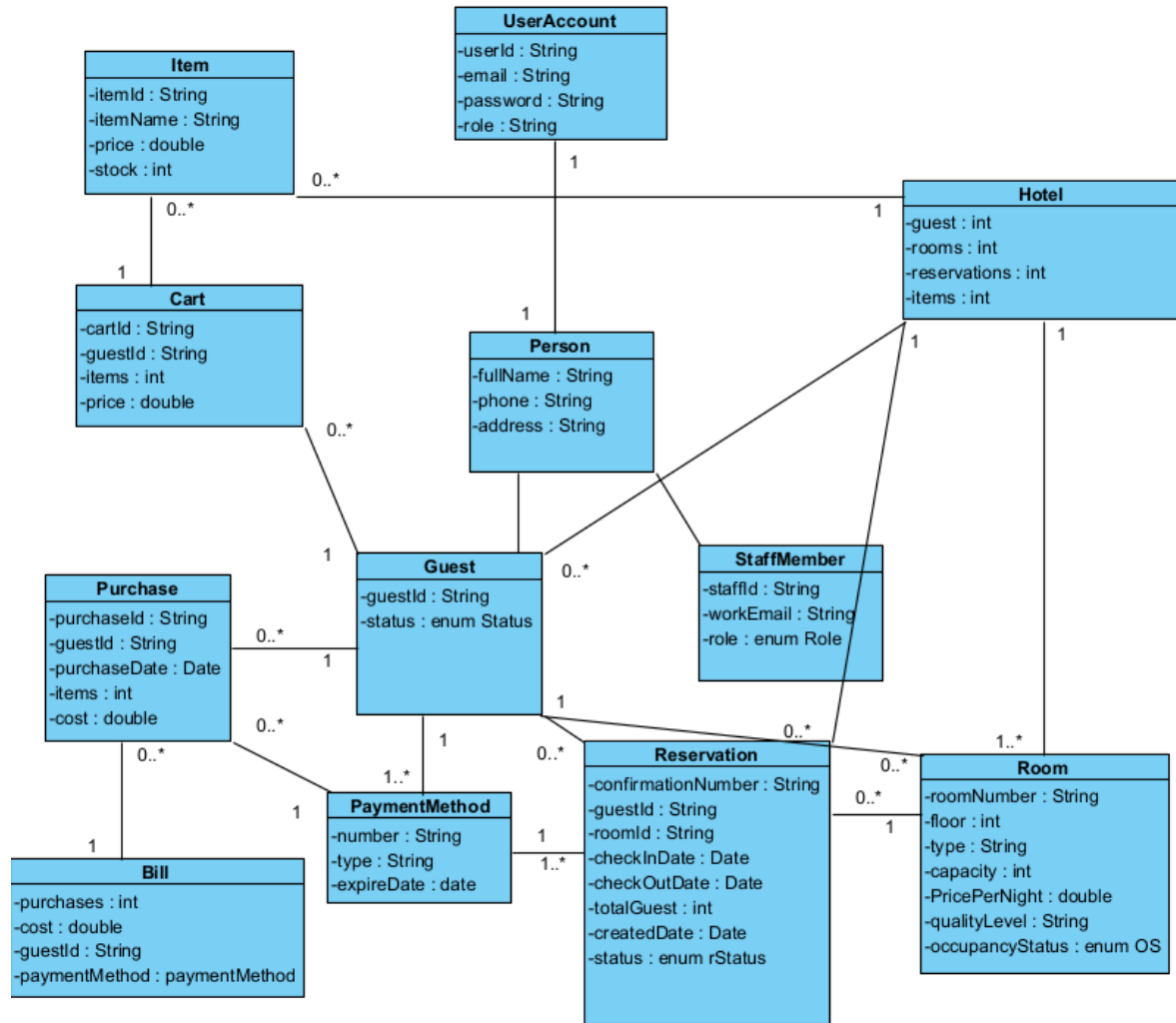


Domain Model



SSDs and Operation Contracts

Contract CO1: bookReservation

Operation: bookReservation(reservation)

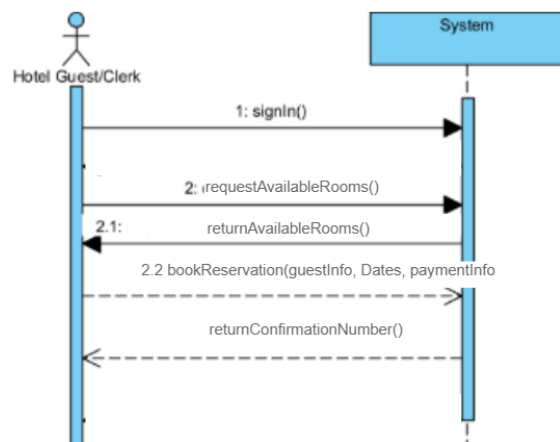
Cross References: Use Case – Sign in, Check in a Guest

Preconditions

1. The system is online
2. Guest is authenticated
3. At least one room = “Available” for requested dates/other details.

Postconditions

1. The room is selected and moved to “reserved/booked/occupied”
2. The reservation is aligned with the correct guest and room.
3. The confirmation number is generated.
4. The system updates the reservation with details.

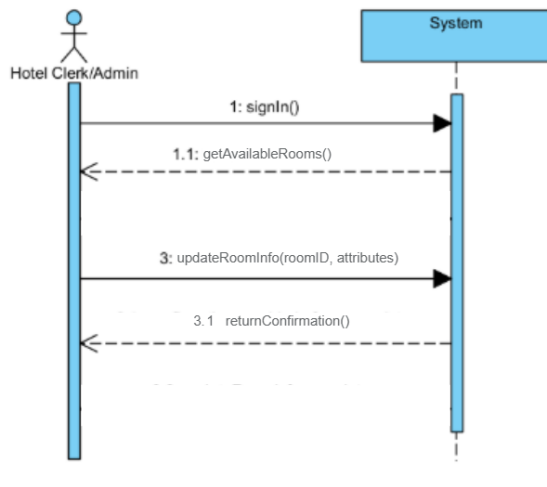


Contract CO1: updateRoomInfo**Operation: updateRoomInfo(room)****Cross References: Use Case – Sign in (clerk, admin)****Preconditions**

4. The system is online.
5. Clerk/Admin authenticated.
6. Room to be selected exists in the system.

Postconditions

5. The room's stored attributes are updated to the new values.
6. The system updates the new room details.

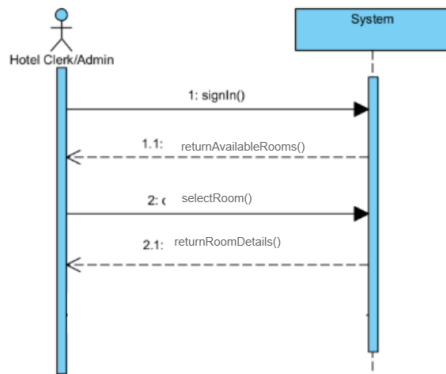


Contract CO1: monitorRoomStatus**Operation: monitorRoomStatus(room)****Cross References: Use Case – Sign in (clerk, admin)****Preconditions**

7. The system is online.
8. Clerk/Admin authenticated.
9. Room exists in the system.

Postconditions

7. The system retrieves the room record.
8. The system returns the status of the room (occupied, reserved, vacant).



Contract CO1: modifyReservation

Operation: modifyReservation(reservationId, newDate)

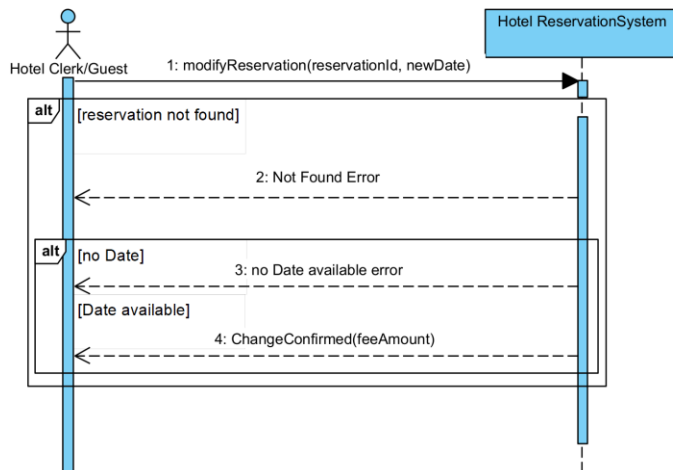
Cross References: Use Case – Sign in (clerk, guest)

Preconditions

1. The system is online and operational.
2. The actor is authenticated or otherwise uniquely identified by the system.
3. The requested stay details are valid (e.g., check-in date before check-out date).
4. Availability data exists (current reservations are accessible for availability checking).

Postconditions

1. A Reservation is created and stored.
2. It records the requested dates and is marked Booked.
3. It is associated with the Guest, selected RoomType, and a payment method.
4. A specific room is assigned (RoomType-level reservation).
5. The system returns a reservationId
6. The system marks the previous reservation as available

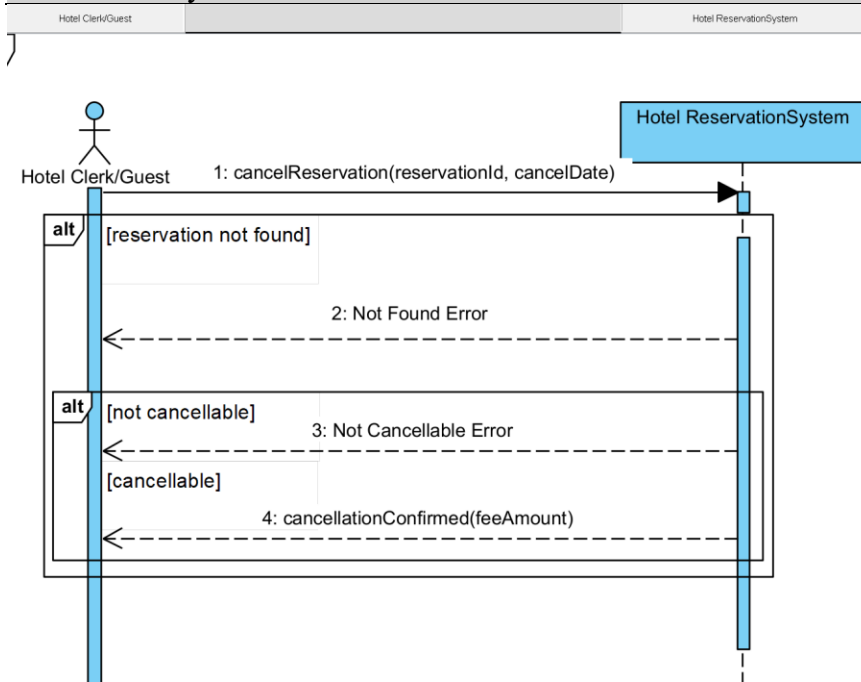


Contract CO1: cancelReservation**Operation: cancelReservation(reservationId, Date)****Cross References: Use Case – Sign in (clerk, guest)****Preconditions**

5. The system is online and operational.
6. The actor is authenticated or otherwise uniquely identified by the system.
7. The requested stay details are valid (e.g., check-in date before check-out date).
8. Availability data exists (current reservations are accessible for availability checking).

Postconditions

7. A Reservation is created and stored.
8. It records the requested dates and is marked Booked.
9. It is associated with the Guest, selected RoomType, and a payment method.
10. No specific room is assigned yet (RoomType-level reservation).
11. The system returns a reservationId



Contract CO1: addGuest

Operation: addGuest(room)

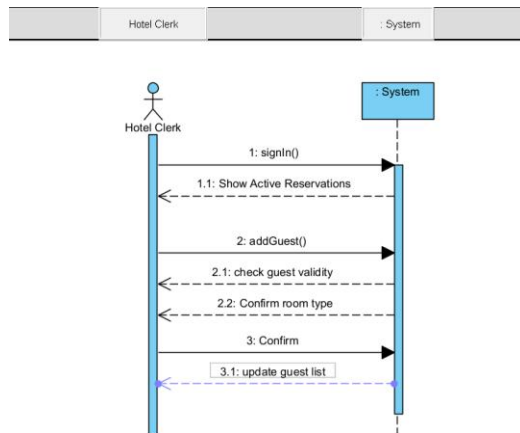
Cross References: Use Case – Sign in (clerk, guest)

Preconditions

10. The system is online.
11. Clerk/Guests authenticated.
12. Room to be selected is valid room type
13. Guest to be added is valid user

Postconditions

9. The guest is added to the reservation



Contract CO1: checkIn

Operation: checkIn(reservation)

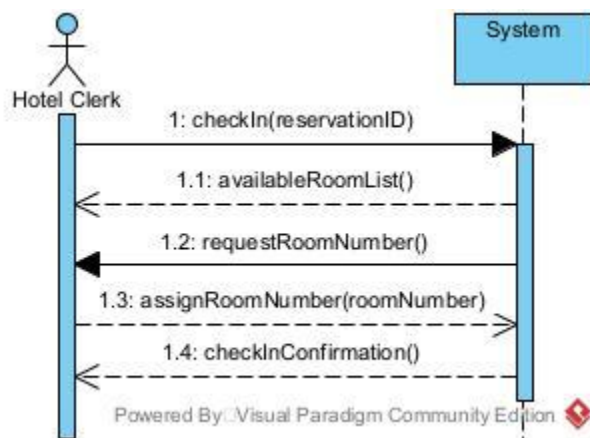
Cross References: Use Case – Check in a Guest

Preconditions

1. The system is online
2. Room and reservation data exist in the system
3. Clerk is logged in
4. Guest has a reservation beginning today

Postconditions

1. Stay object is created, assigned to a room number, and associated with existing Reservation
2. The Reservation's status is updated to "checked-in"
3. The Room is set to "occupied"



Contract CO1: checkOut

Operation: checkOut(reservationID)

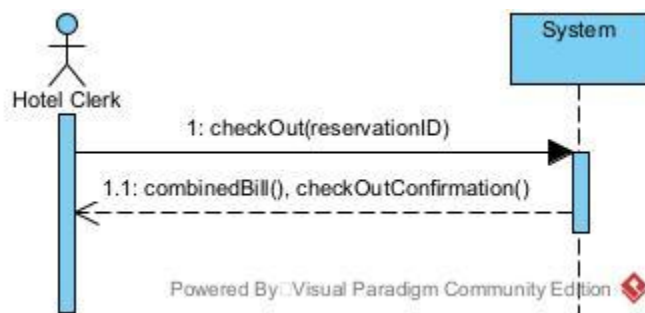
Cross References: Use Case – Check out a Guest

Preconditions

1. The system is online
2. Reservation and Stay objects exist in the system
3. Today is the check-out date, or possible sooner for early check-out
4. Clerk is logged in

Postconditions

1. Total bill is returned
 2. Stay and Reservation are set to “checked-out”
 3. The Room is set to “cleaning”
-



Contract CO1: searchReservations

Operation: searchReservations(checkInDate, checkOutDate, roomTypeID)

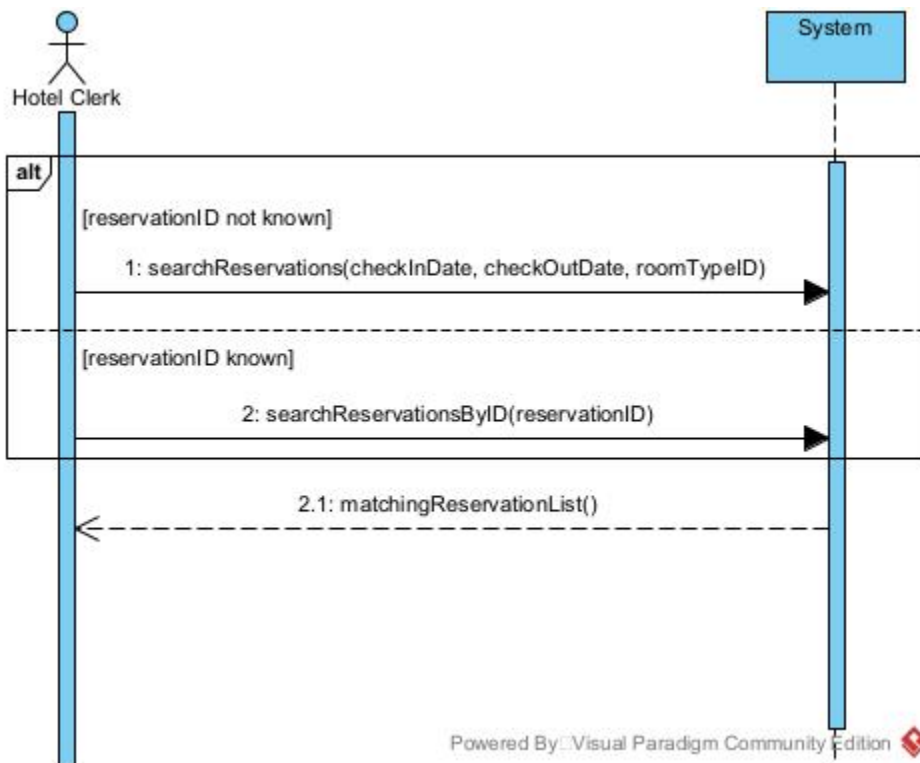
Cross References: Use Case – Search Reservations

Preconditions

1. The system is online
2. Reservation data exists in the system
3. Clerk is logged in

Postconditions

1. A list of Reservations matching the search criteria is returned
2. No Reservation objects are created or modified



Contract CO1: resetPassword

Operation: resetPassword(newPassword)

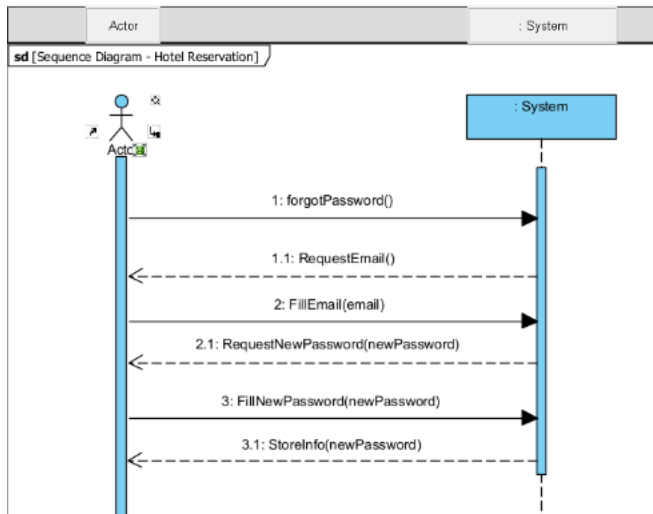
Cross References: Use Case – Reset Password

Preconditions

1. System is online
2. User has a valid account

Postconditions

1. Original User password is updated with new User password.
2. System stores information to user's account.
3. System sends email confirmation that password was updated

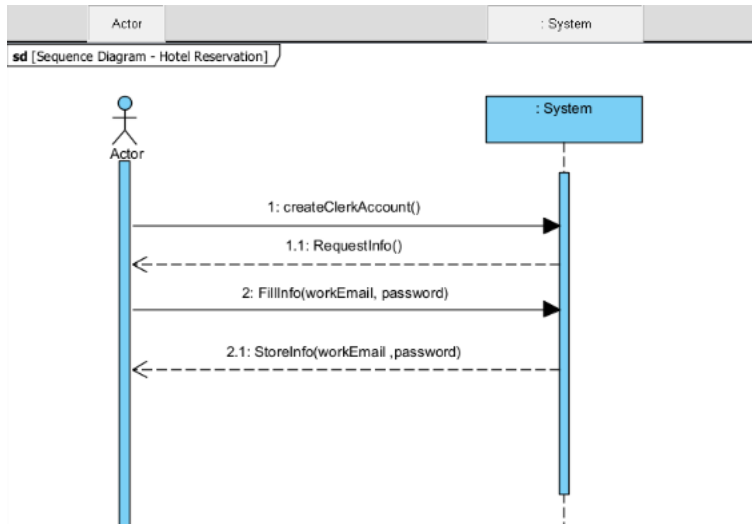


Contract CO1: createClerkAccount**Operation: createClerkAccount(username, password)****Cross References: Use Case – Create Clerk Account****Preconditions**

1. Clerk does not already have an account
2. System is online

Postconditions

1. A Clerk account is created.
2. Systems stores inputted username and password to the Clerk Account.
3. User is able to log in to Clerk Account with their respective credentials.



Contract C01: logOut()

Operation: logOut()

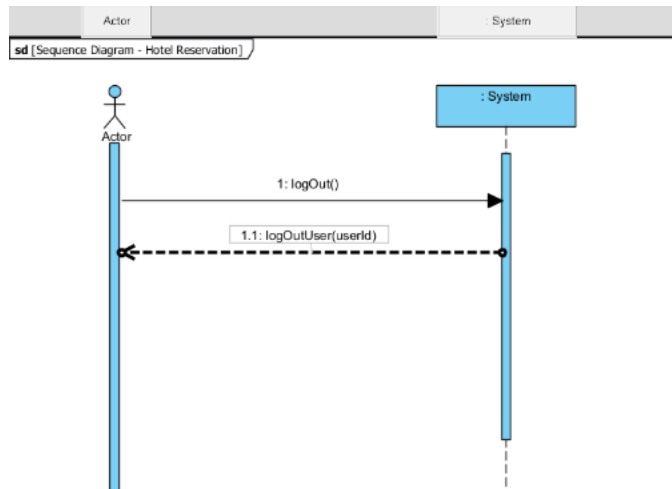
Cross References: Use Case – Log In, Log Out

Preconditions

1. User is logged in their account
2. System is online

Postconditions

1. User account is set to log out.
2. User loses access to account information until logged back in.



Contract CO1: logUserIn

Operation: logUserIn

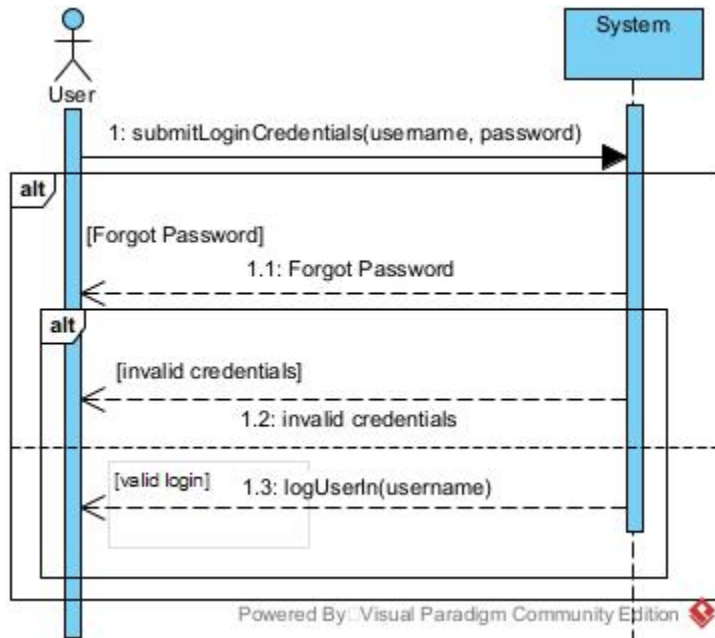
Cross References: Use Case - Login

Preconditions

3. The system is online
4. Login page exists and is available

Postconditions

1. The user is authenticated.



Contract CO2: createAccount

Operation: createAccount

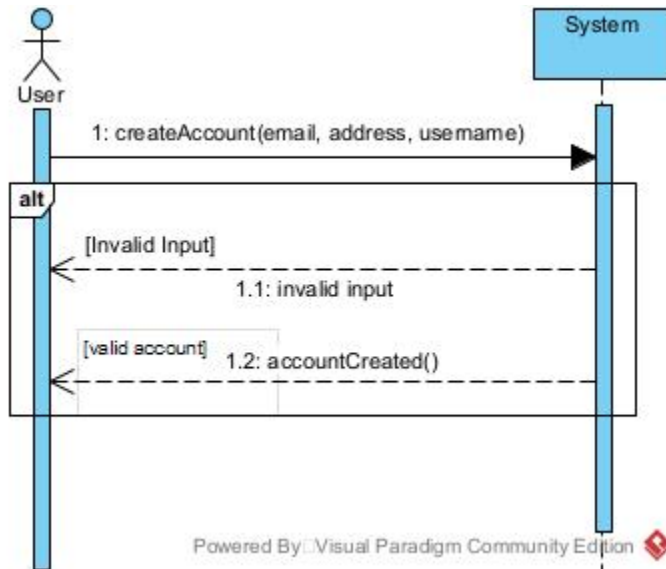
Cross References: Use Case – Create Account

Preconditions:

1. User does not have an existing account.
2. The system is online and operational.

Postconditions

1. User account is successfully created and stored by system



Contract CO1: purchaseItems()

Operation: purchaseItems()

Cross References: Use Case – purchase Items and Generate Combined Bill

Preconditions

1. The system is online
2. The user is an active guest at the hotel
3. The user has an account already

Postconditions

1. The items are recorded as purchased in the system
2. The user's bill is updated to include the purchased items

